

Snort RAG PDF Upload Test - Evaluation Context

Project-specific document for testing upload of evidence and validation context.

Document role	Add controlled evaluation terminology and evidence descriptions for dashboard retrieval tests.
Project	Snort RAG Rule Generator - NLP + Retrieval-Augmented Generation for defensive Snort rule generation.
Use in dashboard	Upload this PDF in the PDF Upload tab to test temporary in-memory knowledge-base extension.
Version	Version 3 - Evaluation Context
Generated	2026-06-09

1. Project evaluation evidence context

This document is not an external dataset. It is a project-specific testing document for the PDF Upload tab. It summarizes the evidence categories expected in the Snort RAG Rule Generator evaluation pipeline.

- The project should demonstrate RAG retrieval, controlled prompt construction, Snort-like rule generation, syntax validation, false-positive checks, and log or PCAP-based evaluation.
- LLM outputs must be interpreted after parsing, validation, repair, and fallback where needed. They should not be described as raw perfect LLM output.
- Controlled real-lab logs are acceptable when clearly documented as lab evidence, not production enterprise traffic.

2. Evaluation records for retrieval and validation

Evidence item	Expected project proof
LLM benchmark	Compare local models such as ollama:qwen2.5, ollama:mistral, and ollama:llama3.2.
Embedding benchmark	Compare BM25, TF-IDF dense retrieval, hybrid retrieval, and reranking.
Snort validation	Show that generated rules pass parser or runtime validation after controlled repair when required.
PCAP replay	Replay generated lab PCAPs and confirm attack categories trigger expected alerts.
Benign traffic	Confirm benign traffic produces zero alerts or NO_RULE_RECOMMENDED.
Real-lab log integration	Use controlled logs to demonstrate classification and rule-selection behavior.

3. Controlled log examples

Log line	Expected label
192.168.10.5 -> 10.0.0.8 HTTP GET /search?q=<script>alert(1)</script>	xss
192.168.10.6 -> 10.0.0.8 HTTP GET /item?id=1 UNION SELECT username,password	sql_injection
192.168.10.7 -> 10.0.0.8 HTTP GET /download?file=../../../../../etc/passwd	directory_traversal
192.168.10.9 -> 8.8.8.8 DNS query example.com and HTTPS browsing	benign_traffic

4. Dashboard behavior expected after upload

- The PDF Upload tab should report the uploaded file name, number of chunks added, new temporary KB size, and first chunk preview.
- The uploaded chunks should only extend the active session knowledge base. They should not overwrite the curated personal dataset.
- Relevant queries should retrieve chunks from this PDF when the query contains matching evidence or attack-family terminology.
- The report screenshot should show that PDF upload supports adding a new document as an extra knowledge source.

5. Safe academic wording

Recommended wording for the final report: The dashboard supports temporary PDF-based KB extension. During testing, a project-specific PDF was uploaded and chunked into the active in-memory knowledge base. The upload function is intended for controlled academic demonstrations and does not replace the main personal dataset.